

Topic 4 Review [99 marks]

1. [Maximum mark: 6]

25M.1.SL.TZ2.5

Construct a trace table for the following algorithm:

```
N = 5
S = 0
R = 0
loop while N > 0
  A = N mod 3
  if A = 0
    then
      S = S - N
    else
      if A = 1
        then
          S = S + N
        else
          S = S + 1
        end if
      end if
      R = R + S
      N = N - 1
    end loop
  output ('The result is ', R)
```

[6]

2. [Maximum mark: 15]

25M.1.SL.TZ2.11

There are 200 students in a school. Their names are held in the one-dimensional string array **STUDENTS**.

The one-dimensional integer array **MARKS** stores marks (0–100 inclusive) that students scored in an examination.

Figure 1: Example data stored in the two arrays **STUDENTS and **MARKS****

STUDENTS		MARKS	
[0]	Ximena Alba	[0]	5
[1]	Boris Mount	[1]	88
[2]	Hugh Parr	[2]	45
[3]	Arjinder Singh	[3]	75
[4]	Mei Chen	[4]	59
	...		...
[199]	Tammy Teller	[199]	51

The one-dimensional integer array **GRADES** will be used to store the grades awarded to students based on their examination marks.

In **Figure 1**, Boris Mount scored 88 marks. His grade will be stored in **GRADES[1]**.

Consider the following algorithm that the school currently uses for awarding grades:

```

loop K from 0 to 199
  GRADES[K] = 1 + (MARKS[K] div 10)
  if GRADES[K] >= 7 then
    GRADES[K] = 7
  end if
end loop

```

(a.i) Determine the value of **GRADES[0]**. [1]

(a.ii) Determine the value of **GRADES[1]**. [1]

(a.iii) Determine the value of **GRADES[2]**. [1]

(a.iv) State the minimum mark necessary to achieve Grade 7. [1]

Grade 1 is a failing grade. Each student who receives a failing grade must re-sit the examination.

- (b) Construct an algorithm in pseudocode to fill the one-dimensional string array **RESIT** with the names of students who must re-sit the examination. [4]

A different method of awarding grades is proposed for the examinations. This new grading system will use three grades represented by the letters A, B, and C.

Grades A, B, or C will be calculated as follows:

- The average mark for all students is calculated.
- Grade A is awarded if an individual student's marks are more than 20 marks above the average mark for all students.
- Grade B is awarded if an individual student's marks are within 20 marks of the average mark for all students.
- Grade C is awarded if an individual student's marks are more than 20 marks below the average mark for all students.

For example, if the average mark for all students is 49.5:

- Grade A is awarded if a student's marks are greater than 69.5.
- Grade B is awarded if a student's marks are in the range from 29.5 to 69.5 inclusive.
- Grade C is awarded if a student's marks are less than 29.5.

- (c) Construct an algorithm in pseudocode to calculate and store the letter grades of all students in the one-dimensional string array **LETTERGRADES** as described. [7]

3. [Maximum mark: 15] 25M.1.HL.TZ2.13

A team of four high-school students decided to create a computer program for younger students to help them learn to count and recognize whole numbers from 1 to 25.

A table partially completed with numbers from 1 to 25 is given to a younger student. The table consists of five rows and five columns (see **Figure 2**). The

younger student must enter the remaining numbers. Each number from 1 to 25 should be entered only once.

Figure 2: Example table partially filled with numbers

	24	1		
		7	14	
4	6			22
10			21	3
11		25		9

Each of the four students is responsible for creating several sub-programs that will be included in the program.

- (a) List **three** advantages of using sub-programs in this situation. [3]

The sub-program `inputInRange ()` should return a valid input number.

This will ensure that the younger student's input is a number in the range 1–25 inclusive. If the input number is not in this range, an appropriate message is output. The process will be repeated until a valid number is input.

For example:

For the sub-program call `N = inputInRange ()`, if the following appears on the screen:

Enter a number:

 The number is not in the range from 1 to 25. Please try again.
 Enter a number:

 The number is not in the range from 1 to 25. Please try again.
 Enter a number:

 Your number is valid.

Then the value of N would be 15 after execution.

- (b) Construct an algorithm in pseudocode for the sub-program
 inputInRange().

[4]

In the program, the table is stored as a static two-dimensional array.

Figure 3: Two examples of data

Figure 3a: The two-dimensional array is correctly completed because each of the numbers from 1 to 25 appears only once.

	[0]	[1]	[2]	[3]	[4]
[0]	2	24	1	8	15
[1]	23	5	7	14	16
[2]	4	6	13	20	22
[3]	10	12	19	21	3
[4]	11	18	25	17	9

Figure 3b: The two-dimensional array is not correctly completed because the number 17 is missing and the number 12 appears twice.

	[0]	[1]	[2]	[3]	[4]
[0]	12	24	1	8	15
[1]	23	5	7	14	16
[2]	4	6	13	20	22
[3]	10	12	19	21	3
[4]	11	18	25	2	9

The sub-program isCorrectlyCompleted(MAT) accepts the two-dimensional

array **MAT** and outputs an appropriate message to indicate if it has been correctly filled.

For example:

If **MAT** is the two-dimensional array shown in **Figure 3a**, then `isCorrectlyCompleted(MAT)` should output that the table has been correctly completed.

If **MAT** is the two-dimensional array shown in **Figure 3b** then `isCorrectlyCompleted(MAT)` should output that the table has **not** been correctly completed.

The description of the algorithm for the `isCorrectlyCompleted(MAT)` sub-program is as follows:

- Initialize a one-dimensional array, **FLAGS**, with zero (0) values.
- Visit all elements of the **MAT** array.
 - Use the number stored in the **MAT** array to match up the index in the **FLAGS** array.
 - Assign one (1) to the **FLAGS** array at that index.
- Inspect elements of the **FLAGS** array.
 - If all the elements in the **FLAGS** array are one (1), output a message saying that the table has been completed correctly; otherwise, output a message saying that the table has not been completed correctly.

For example, the **FLAGS** array for the **MAT** array shown in **Figure 3a** is:

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]	[22]	[23]	[24]
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

And the **FLAGS** array for the **MAT** array shown in **Figure 3b** is:

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]	[22]	[23]	[24]
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1

- (c) Construct the algorithm in pseudocode for the `isCorrectlyCompleted(MAT)` sub-program as described. [8]

4. [Maximum mark: 4] 24N.1.SL.TZ0.9
Construct a trace table for the following algorithm:

```
A = 20
B = 12
loop while B > 0
  TEMP = B
  B = A mod B
  A = TEMP
end loop
output (A)
```

[4]

5. [Maximum mark: 2] 24N.1.SL.TZ0.10
State **two** characteristics of a collection. [2]

6. [Maximum mark: 15] 24N.1.SL.TZ0.13
Mean and median are different measures of central tendency for a numerical dataset.

Figure 1: Example data values stored in a one-dimensional array, `DATA`, which contains 10 elements

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
2	3	3	6	5	9	10	100	14	17

The mean of the data values stored in a one-dimensional array can be determined as follows:

- Add up all the numbers in the one-dimensional array.
- Divide this sum by how many numbers there are.

The mean of the data values stored in the **DATA** array in **Figure 1** is 16.9.

- (a) Construct an algorithm in pseudocode to calculate and output the mean of the data values stored in the one-dimensional array **DATA** that contains **N** elements. You may assume that the **DATA** array containing **N** elements has been inputted. [4]

The median of the data values stored in a one-dimensional array can be determined as follows:

- Arrange the data values from the lowest to highest value.
- The median is the data value in the middle of the array.
- If there are two data values in the middle, then the median is the mean of those two values.

The median of the data values stored in the **DATA** array in **Figure 1** is 7.5.

- (b.i) State **one** algorithm that could be used to arrange the data values from the lowest to highest value. [1]
- (b.ii) Construct an algorithm in pseudocode to determine and output the median of the data values stored in the one-dimensional array **DATA** that contains **N** elements. You may assume that the **DATA** array containing **N** elements has been inputted. [10]

7. [Maximum mark: 2] 24N.1.HL.TZ0.6
State **two** benefits of using subprograms within a computer program. [2]

8. [Maximum mark: 15] 24M.1.SL.TZ1.15
(a) Describe **one** standard operation of collections. [2]

The collection **RAINBOW** holds the names of the main colours of the rainbow along with their order indicator, as follows:

Blue, E, Green, D, Indigo, F, Orange, B, Red, A, Violet, G, Yellow, C

- (b) Construct an algorithm using pseudocode to read the data from **RAINBOW** and store the names of the colours in a one-dimensional string array, **COLOUR**, and the order indicators in another one-dimensional string array, **ORDER**.

The matching colour and order indicator must have the same index in the **COLOUR** and **ORDER** arrays.

[4]

- (c) Construct an algorithm using pseudocode to sort the contents of the array **ORDER** into alphabetical order. The indexes for the corresponding data in the two parallel arrays must remain the same after sorting.

[5]

- (d) Compare and contrast the bubble sort algorithm and the selection sort algorithm.

[4]

9. [Maximum mark: 15]

24M.1.SL.TZ2.15

- (a) Describe **one** standard operation of collections.

[2]

The collection **CAPITALS** holds the names of a number of capital cities and the names of their corresponding countries, as follows:

Ankara, Turkey, Brasilia, Brazil, Dhaka, Bangladesh, Lisbon, Portugal, Manila, Philippines, Rome, Italy

- (b) Construct an algorithm using pseudocode to read the data from **CAPITALS** and store the names of the cities in a one-dimensional string array, **CITY**, and the names of the countries in another one-dimensional string array, **COUNTRY**.

- The matching city and country must have the same index in the CITY and COUNTRY arrays. [4]
- (c) Construct an algorithm using pseudocode to sort the contents of the array COUNTRY into alphabetical order. The indexes for the corresponding data in the two parallel arrays must remain the same after sorting. [5]
- (d) Compare and contrast the bubble sort algorithm and the selection sort algorithm. [4]
- 10.** [Maximum mark: 1] 23N.1.SL.TZ0.2
Define the term *constant*. [1]
- 11.** [Maximum mark: 3] 23N.1.SL.TZ0.8
(a) Outline what is meant by concurrent processing. [2]
(b) Identify **one** advantage of concurrent processing. [1]
- 12.** [Maximum mark: 6] 23N.1.SL.TZ0.9
(a) Construct a trace table for the following algorithm:
 $N = 1216$
 $X = 0$
loop while $N > 0$
 $X = X + N \bmod 10$
 $N = N \text{ div } 10$
end loop
output (X) [4]
(b) Deduce the purpose of the algorithm. [2]

